

Ethanol Blending in Gasoline – Croatia

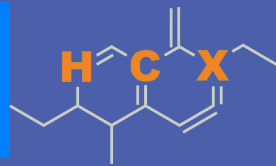
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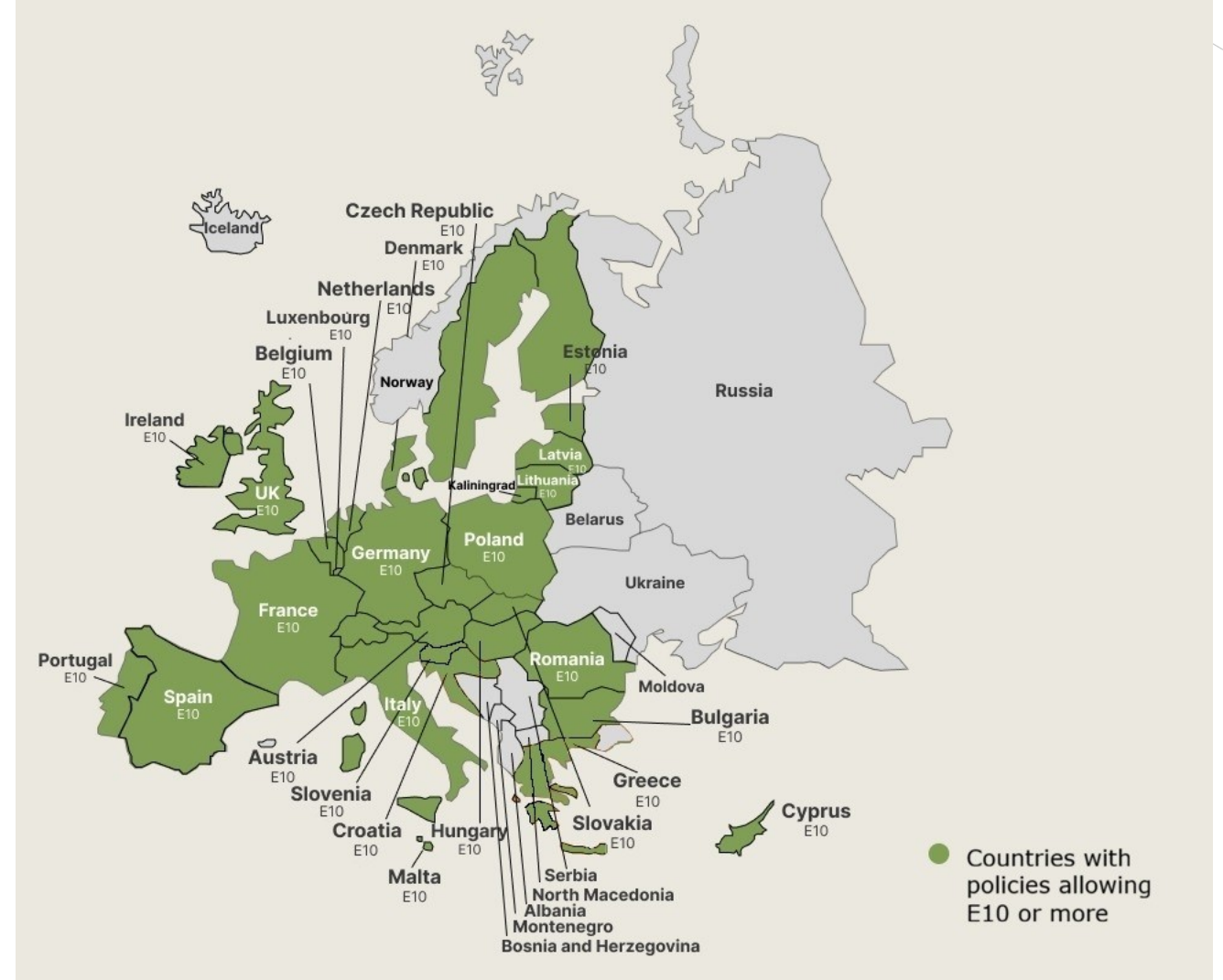


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

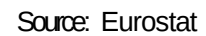
Ethanol Gasoline Blending in Croatia



In 2025, gasoline consumption in Croatia was 772 million liters (204 million gallons) and growth rate was -0.8%. Gasoline production and net imports were 1,310 and -510 million liters (346 and -135 million gallons) correspondingly. Croatia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

Although E10 is allowed in Croatia, it does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups at the 3 and 5 positions.

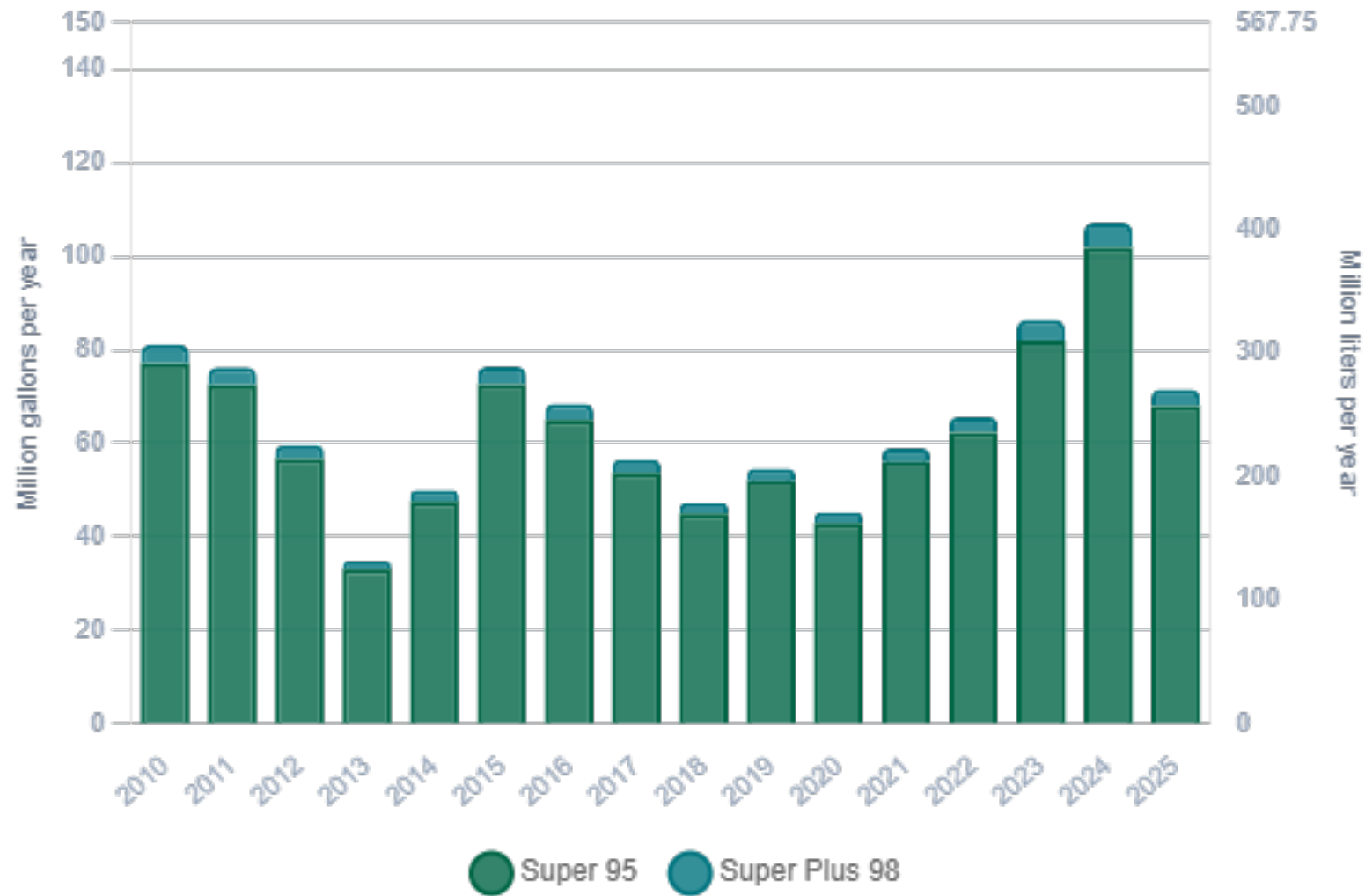


Gasoline Production in Croatia



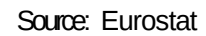
Source: Eurostat

Gasoline Imports in Croatia

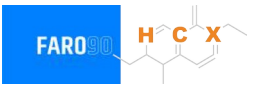


Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a methyl group, a propyl group, and a group labeled 'X' which is connected to a chain containing a carbonyl group and a methyl group. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.



Ethanol Balance in Croatia



Croatia does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

Source: The Global Economy, WITS/Comtrade, Total Croatia News, Axens

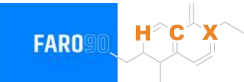
Gasoline Quality in Croatia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.8							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	1.0							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

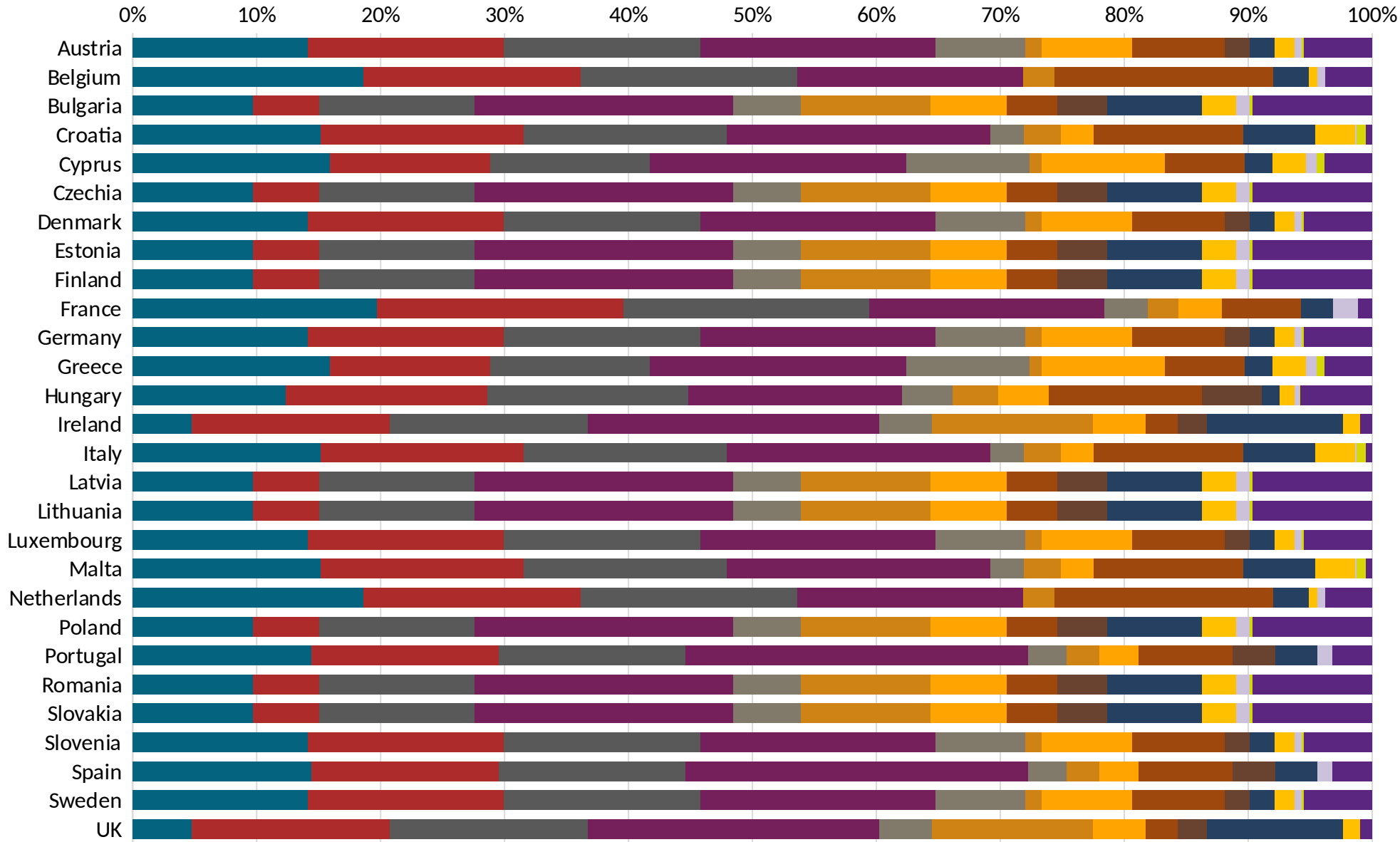
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



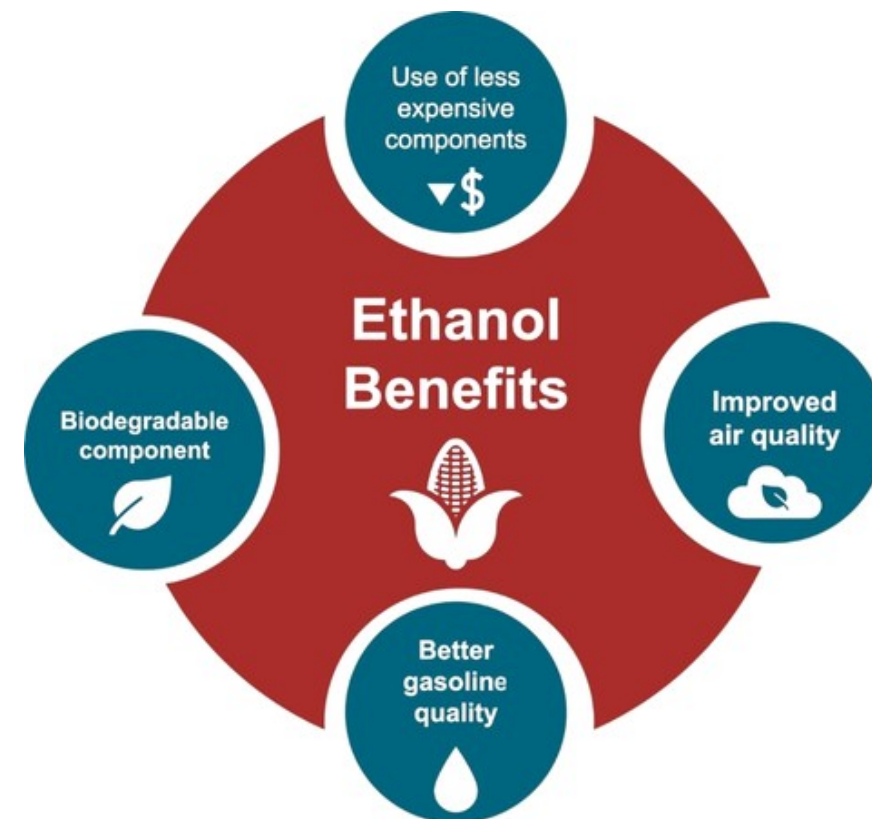
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

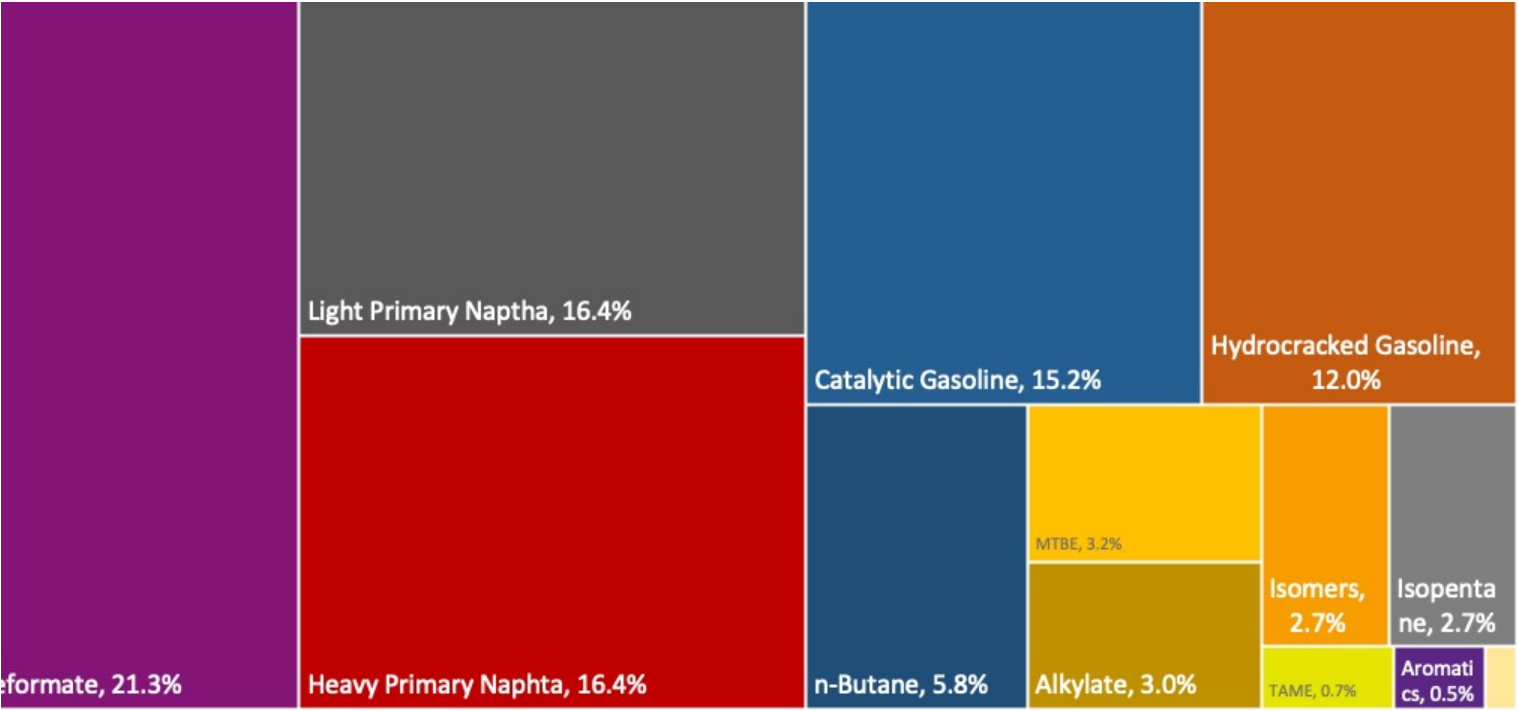
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

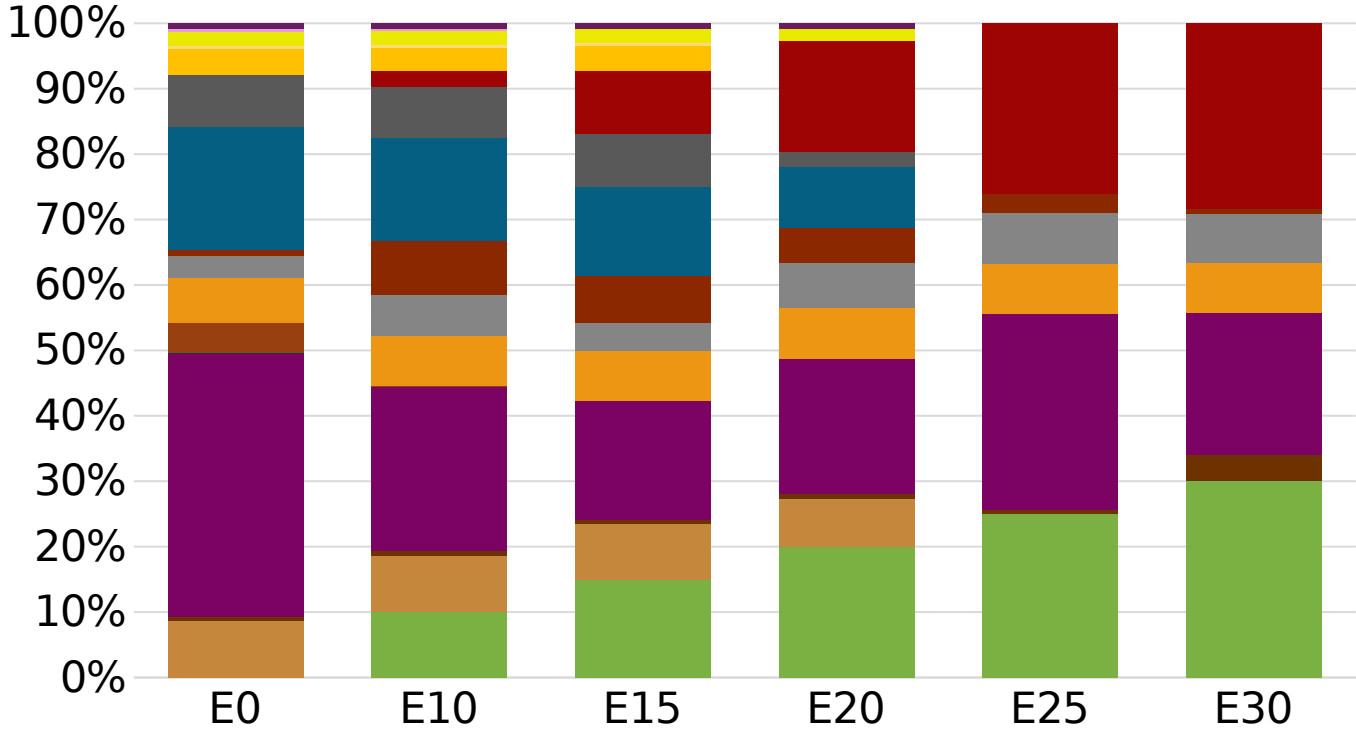
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Croatia Available Blending Components



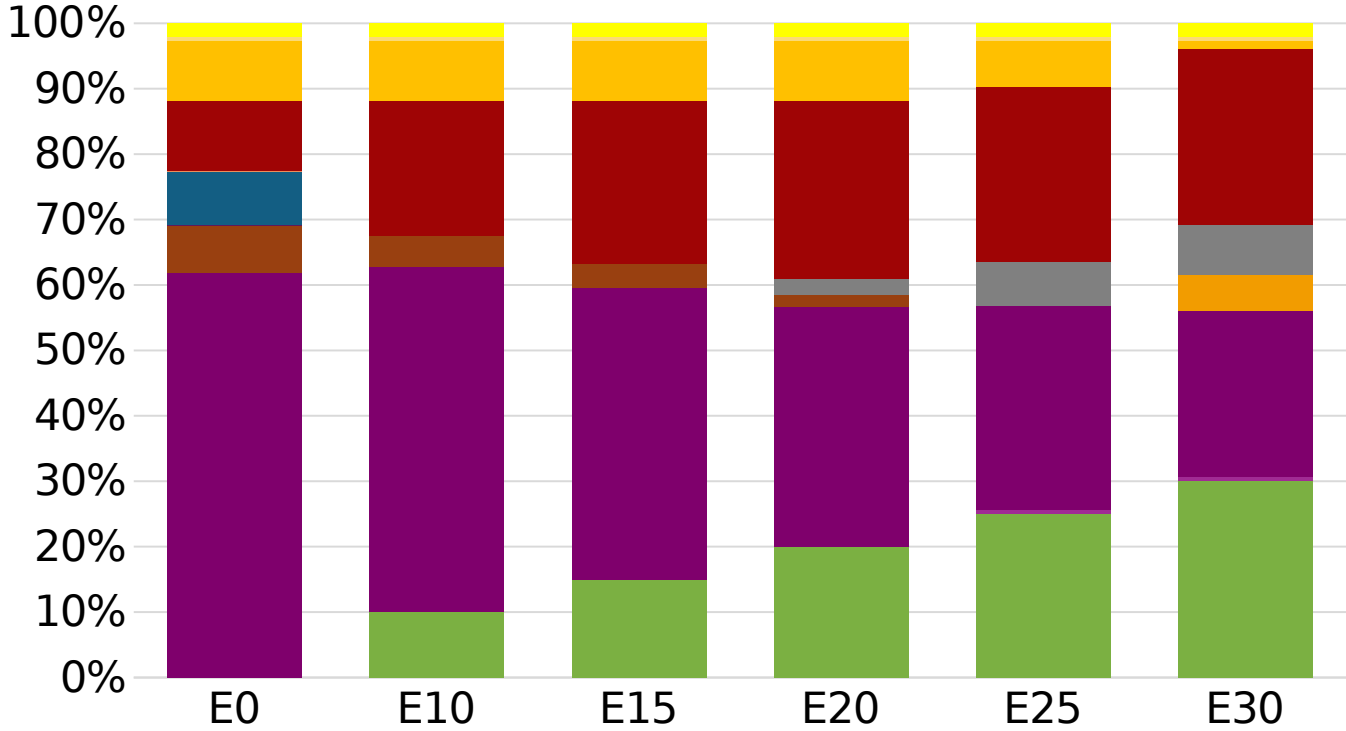
Croatia – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.055	\$ 1.942	\$ 1.859	\$ 1.783	\$ 1.688	\$ 1.614

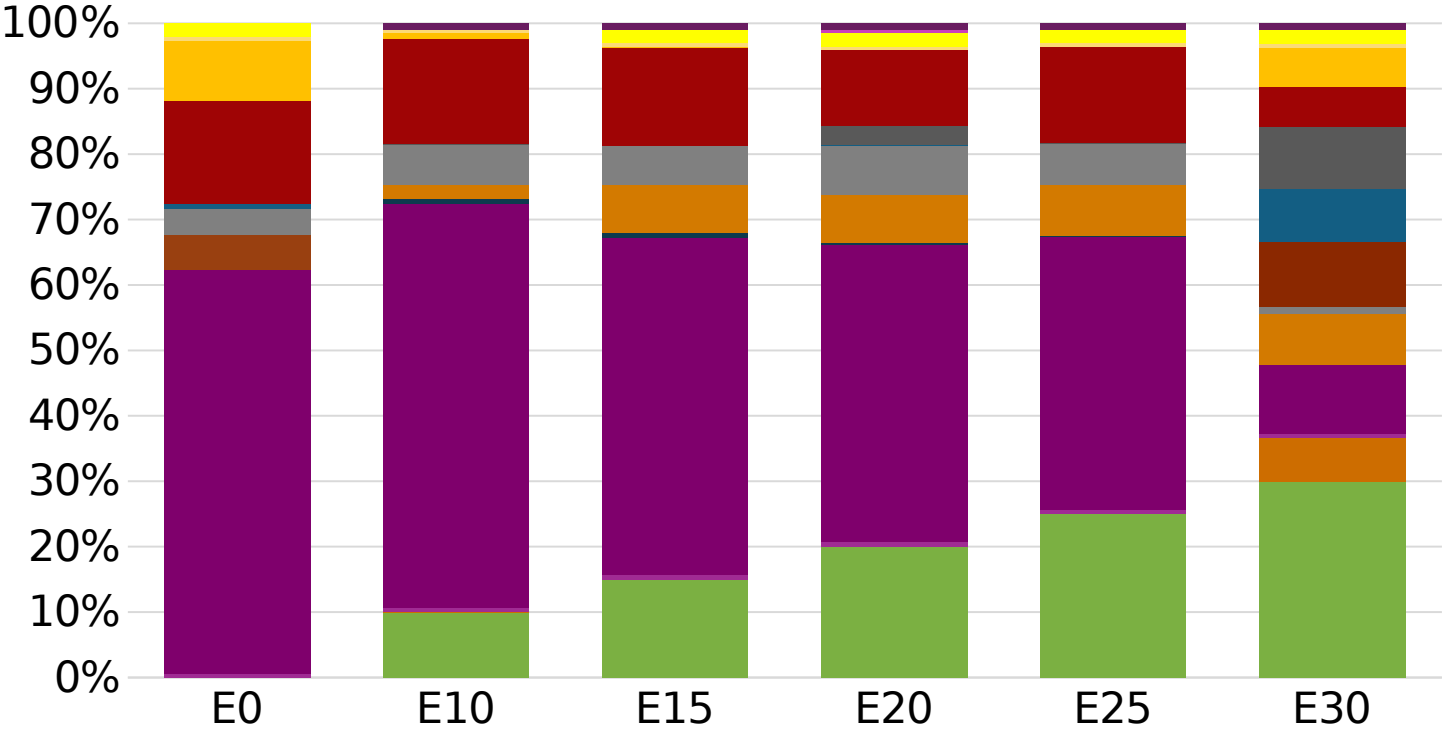
Croatia - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.104	\$ 1.985	\$ 1.916	\$ 1.846	\$ 1.776	\$ 1.723

Average CIF 2025
Prices

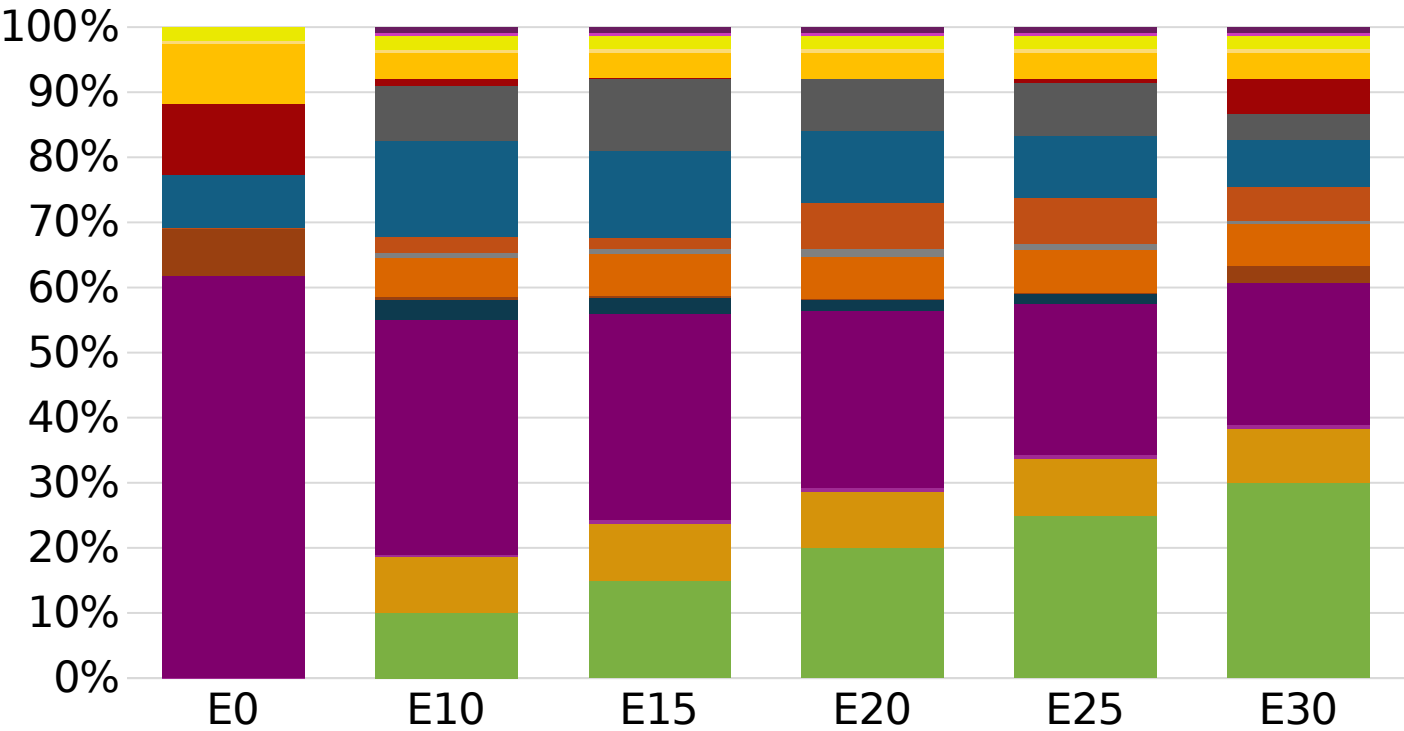
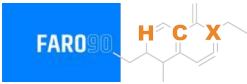
Croatia – Gasoline 95 – Increased Octane



Average CIF 2025 Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974

Croatia – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104

Source:
Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

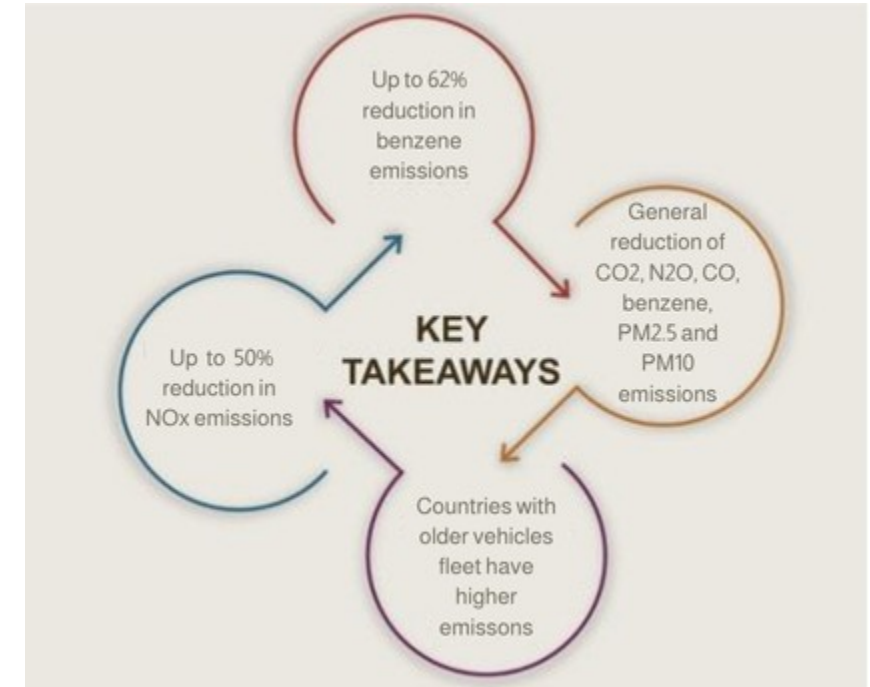
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

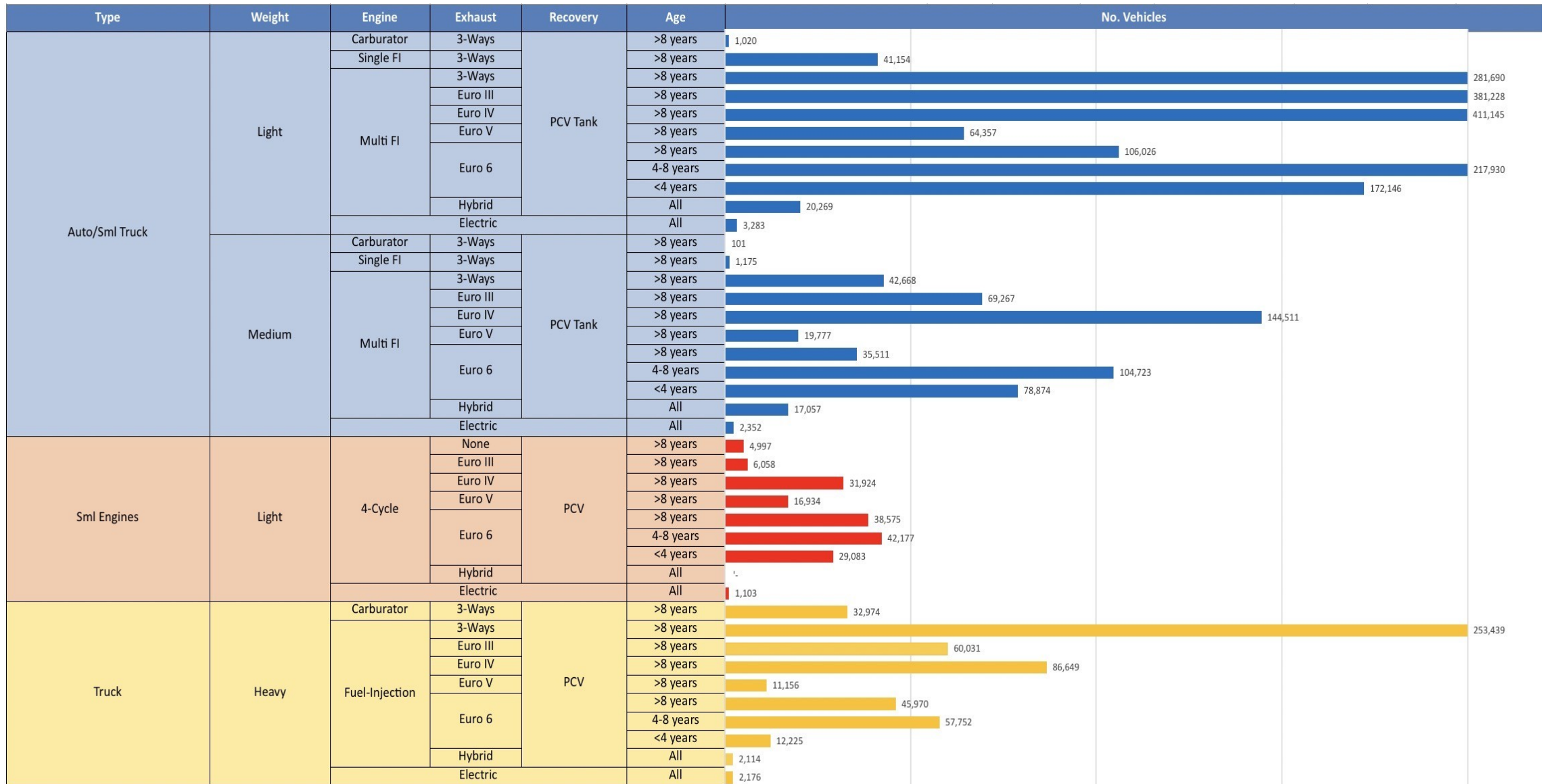
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Croatia

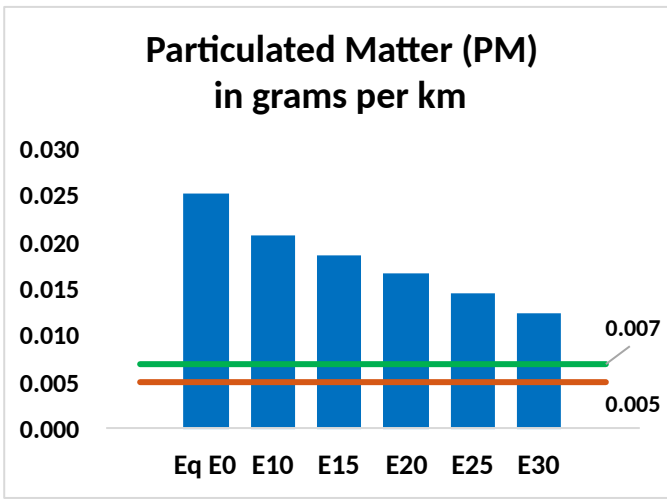
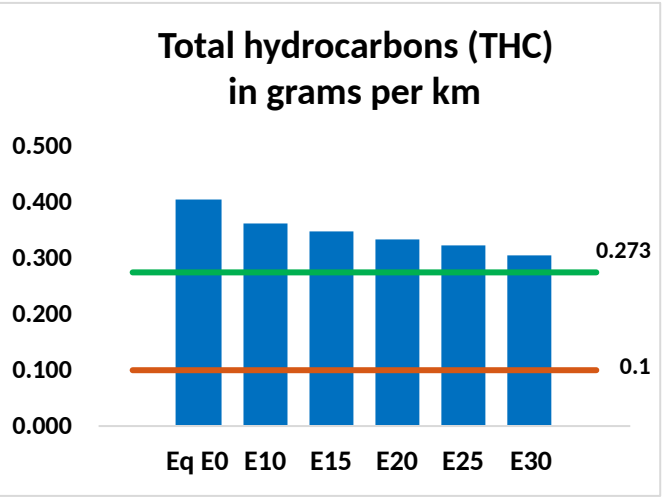
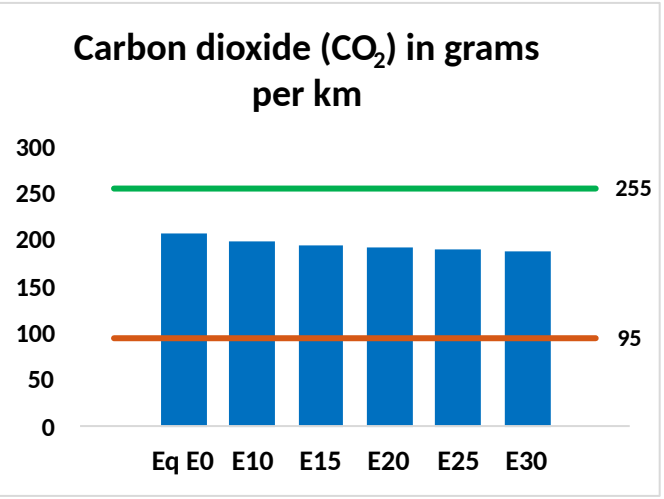
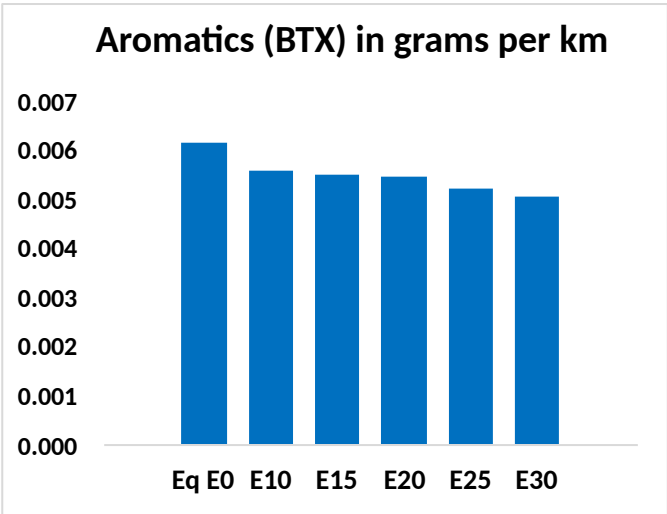
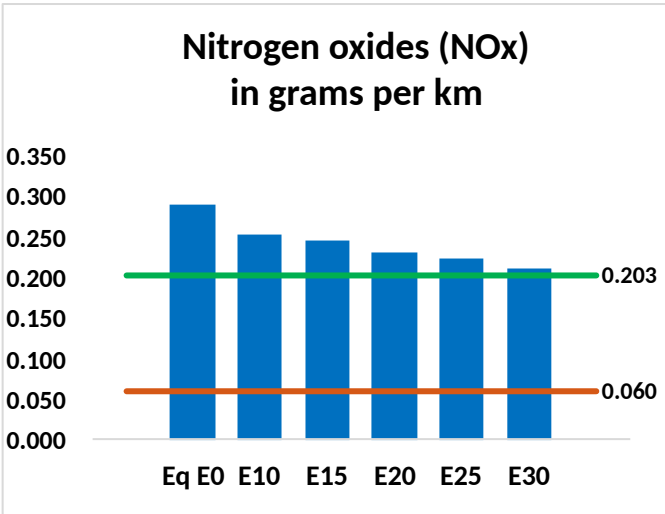
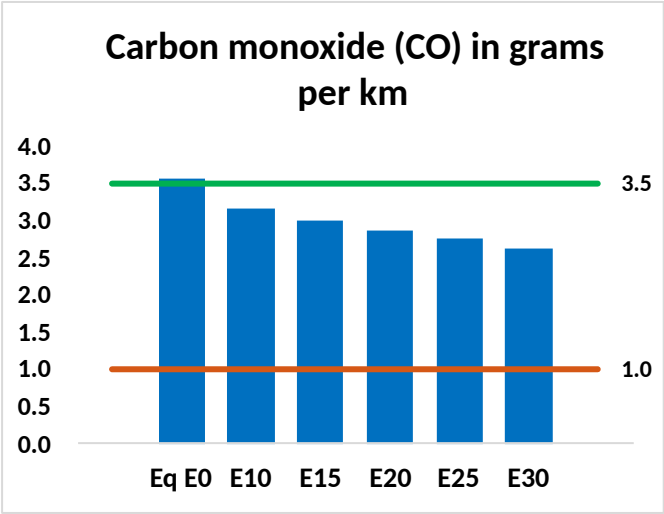
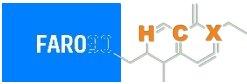


Vehicle Fleet: 2,951,603
Gasoline Vehicle Fleet: 909,203

Source: UNECE, Eurostat, ACEA

Croatia – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Croatia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	86,529	81,621	76,713	73,032	69,483	66,802	63,456	-11%	-20%
CO2	5,060,220	4,930,100	4,807,209	4,710,559	4,662,733	4,634,205	4,548,784	-5%	-8%
VOC	9,859	9,341	8,824	8,446	8,088	7,820	7,438	-10%	-18%
NOx	7,061	6,520	6,150	5,963	5,660	5,423	5,144	-13%	-20%
SOx	57	53	49	45	42	38	34	-15%	-28%
NH3	1,754	1,737	1,719	1,727	1,747	1,760	1,772	-2%	0%
N2O	71	71	71	71	72	73	74	-1%	2%
CH4	2,170	2,170	2,170	2,203	2,250	2,292	2,331	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	85	82	80	79	79	78	77	-5%	-7%
Acetaldehyde	200	230	338	514	700	800	946	68%	249%
Formaldehyde	657	639	745	875	917	1,014	1,104	13%	39%
Benzene	150	144	136	134	133	127	123	-9%	-11%
PM2.5	300	273	246	222	198	173	147	-18%	-34%
PM10	315	287	258	233	208	182	154	-18%	-34%

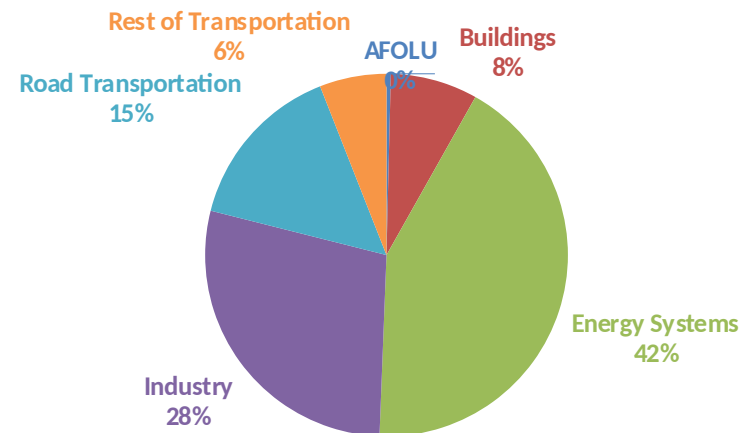
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

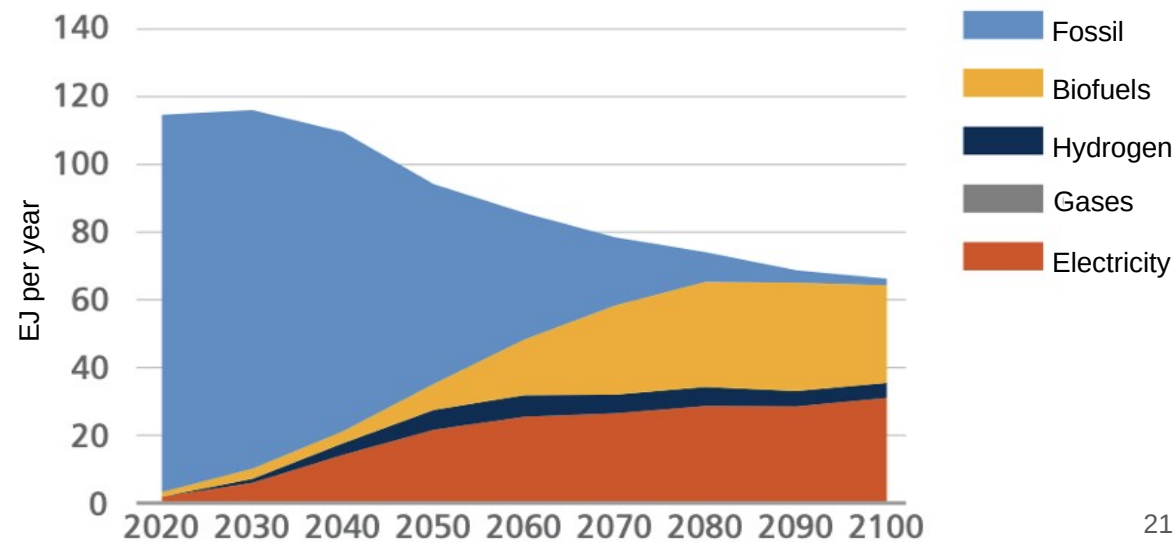
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

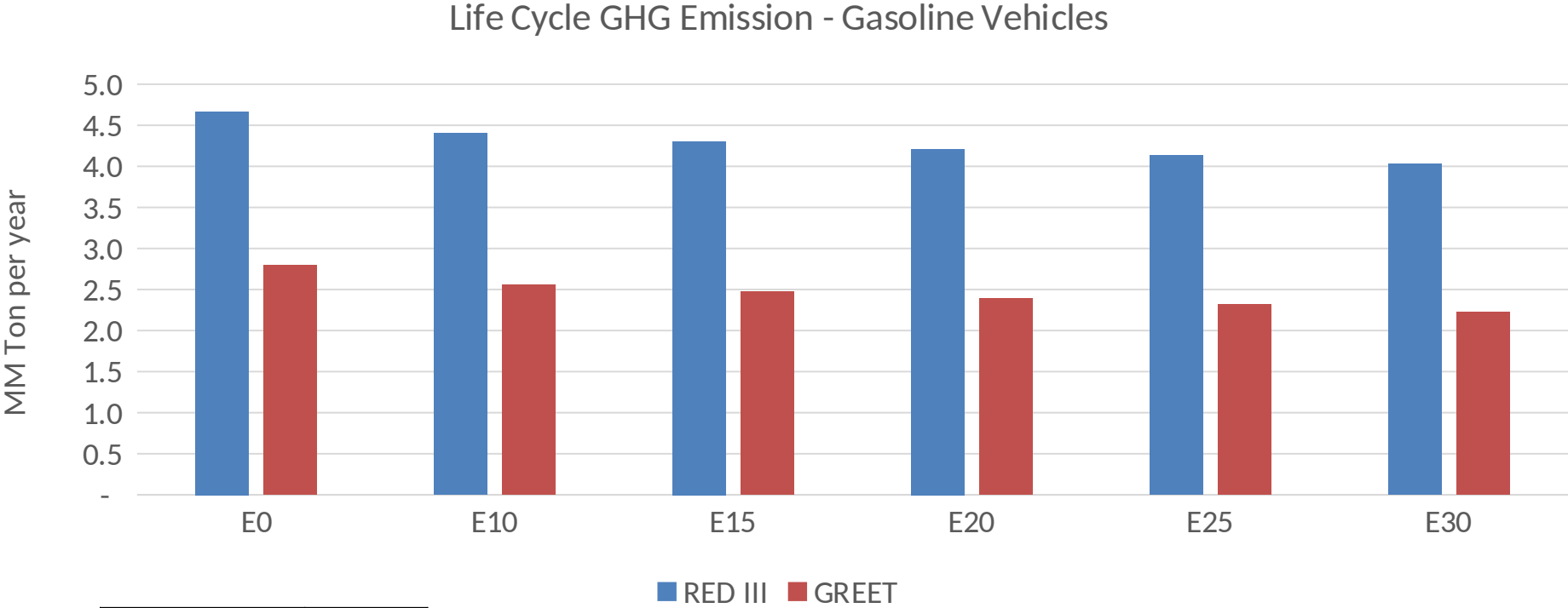
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Croatia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	17.9
Target @ 2035 (MMT/year)	10.1
Delta	7.8

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90